Experiment 11: Top-Down Parser for Context-Free Grammar in Python

1. Aim

To write a Python program to implement a simple top-down parser for context-free grammars.

2. Procedure

1. Import `CFG` and `RecursiveDescentParser` from the `nltk` library.
2. Define a context-free grammar using `CFG.fromstring()`.
3. Create a top-down parser object using `RecursiveDescentParser`.
4. Split the input sentence into tokens and pass it to `parser.parse()`.
5. Display the generated top-down parse tree.

3. Program

```
import nltk
from nltk import CFG

grammar = CFG.fromstring("S -> NP VP\nNP -> Det N\nVP -> V NP\nDet -> 'the' | 'a'\nN -> 'cat' | 'rat'\nV -> 'chased'")

parser = nltk.RecursiveDescentParser(grammar)
sentence = "the cat chased a rat".split()

for tree in parser.parse(sentence):
    print("Top-Down Parse Tree:", tree)
```

4. Output

```
Top-Down Parse Tree: (S (NP (Det the) (N cat)) (VP (V chased) (NP (Det a) (N rat))))
```

5. Result

Thus, the Python program was successfully executed to implement a top-down parser for context-free grammars using `RecursiveDescentParser`.